

Average case (Sequential Search)

$i=0 \rightarrow 1$ comparison

$i=1 \rightarrow 2$ comparison

$i=2 \rightarrow 3$ comparison

:

$i=n-1 \Rightarrow n$ comparison.

Now

	No of steps
int Sequential(int a[], int n, x)	0
{	0
int i;	1
for (i=0; i<n && a[i]!=x; i++)	j+1
if (i==n) return -1	1 (comparison)
else return i;	1 (return)
}	<u><u>j+4</u></u>

$$\sum_{j=0}^{n-1} j+4$$

$$\Rightarrow \frac{(n-1)n}{2} + 4$$

$$\frac{n^2 - n + 8}{2} \times \frac{1}{n}$$

$$\frac{n^2 - n + 8}{2n} = \frac{n}{2} - \frac{1}{2} + \frac{4}{n}$$

$$= O(n)$$